MATH11218
Methods and Basically Specialist Revision

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Within the expression of a^m, a is considered to be the base with the power m being the index. The reason that this is considered useful is in scientific notation to expression large numbers, or simplify repeated multiplication, fractions, roots or polynomials. E.g. $2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6$. With these there are a number of different index laws that goes with it:

1. $a^m \cdot a^n = a^{m+n}$
2. $\frac{a^m}{a^n} = a^{m-n}$
3. $(a^m)^n = a^{m \cdot n}$
4. $(ab)^m = a^m b^m$
5. $(\frac{a}{b})^m = \frac{a^m}{b^m}$
6. $a^{\frac{m}{n}} = \sqrt[n]{a^m}$
7. $a^0 = 1$